



Context-dependent Impulsivity in Substance Use Disorder: A Neuroscience-informed Framework

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Abstract

Purpose of Review Due to various reasons including psychobiological states and psychopathological conditions, individuals may be more inclined toward choices or actions that are known as impulsive. This preference arises from either or both personality characteristics (*trait*), which are more stable, or impulsive tendencies that emerge in response to environmental factors and/or internal factors (e.g., mood) and fluctuate over time (*state*). Although both types of impulsivity are closely tied to addiction, particularly in terms of treatment outcomes (e.g., poor adherence, increased relapse risk), state impulsivity is susceptible to influences of context and the saliency of cues. One specific form of state impulsivity, known as context-dependent impulsivity, is evoked by contextual drug-related cues and could be conceptualized, monitored, and assessed using the cue-reactivity paradigm. In this paper, we aim to tease apart the neurocognitive mechanisms which are influenced or heightened by context-dependent impulsivity and explain how these mechanisms may interfere with addiction treatment.

Recent Findings By integrating evidence from classic neuropsychological measures and the drug-cue reactivity paradigm in the field of addiction, we propose a neuroscience-informed conceptual framework explaining how relapse may occur due to the disruption of neurocognitive mechanisms in response to contextual cues. We introduce the term 'context-dependent impulsivity' in accordance with the cue-reactivity paradigm as a marker of addiction recovery and as a mechanistic target for cue-based interventions such as cognitive bias modification and mindfulness-based relapse prevention.

Summary Context-dependent impulsivity is a multi-faceted construct, which could be characterized by specific cognitive phenotypes. Parsing these different phenotypes into their constituent neurocognitive mechanisms and examining their interactions with relapse vulnerability could inform clinicians about optimized cognitive interventions and neuropsychological assessments tailored to individuals with specific impulsivity phenotypes within the trait or state levels.

Keywords Neurocognitive processes · Context-dependent impulsivity · Substance use disorders · Addiction treatment

Introduction

Substance use disorder (SUD) is a major public health problem. According to the National Survey on Drug and Health (NSDUH), in 2022, 46.6 million people aged 12 or older used illicit drugs in the past month [1]. Impulsivity is a factor frequently linked to the onset, progression, and relapsing into the consumption of drugs and alcohol, identified as a cardinal predictor and consequence of drug use behavior [2–5]. Being associated with all stages of the addiction cycle (binge/intoxication, preoccupation/anticipation, withdrawal/negative affect), impulsivity could be targeted for addiction prevention and treatment with preventive/rehabilitative interventions [6].

The term *impulsivity* has been part of human language for more than five centuries and has been acknowledged as

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having a significant role in psychiatric conditions from the very beginning of the fields of neurology and psychiatry [7]. Impulsivity is defined as a tendency for swift, unplanned reactions to internal or external stimuli without forethought as to how these reactions might have negative consequences for oneself or others [8]. Impulsivity could be conceptualized using the Dual Process Model by Daniel Kahneman [3] and others [9, 10]. According to this model, there are two co-existing cognitive systems underlying thinking and reasoning: system 1 (automatic) and system 2 (analytical). Impulsivity arises from the imbalance between these two opposite systems, in a simplified framework [2, 3]. According to Kahneman's theory, system 1 is responsible for rapid, automatic, intuitive, and unconscious processes, while analytical and reflective processes are associated with system 2. Balanced contributions between these two systems lead to efficient self-control and adaptive response to stimulus/expected outcome. Time pressure, psychobiological states (e.g., hunger, anger, fatigue), emotional distress, and psychopathology (e.g., SUD, ADHD) disturb the balance between the two systems and make individuals more reliant on system 1 when making a decision or taking an action. Hyperactivation of system 1, in contrast to hypoactivation of system 2 contributes to weakened inhibition and results in impulsive behavior [7].

Impulsivity is a multidimensional construct that has both stable "trait" dimensions as well as more fluctuating "state" dimensions [11]. Trait impulsivity can be considered as an overall consistent (personality) characteristic with genetic loading that is usually measured with self-report questionnaires such as the Barratt Impulsiveness Scale, the Urgency-Premeditation-Perseverance-Sensation Seeking-Positive Urgency (UPPS-P) impulsive behavior scale [12] and the Eysenck impulsiveness questionnaire [13, 14]. More specifically, in Eysenck's conceptual framework impulsivity gradually became a core feature of the personality trait of *extraversion* and was subsequently considered central to arousal-based theories of cognitive performance [15, 16]. Trait impulsivity has had an important place in the study of human personality since Heymans's three-dimensional model of temperament [17], and to date, impulsivity-related items are included in the majority of self-report questionnaires and personality inventories [18]. Trait impulsivity is typically associated with several heritable tendencies (with significant genetic loading), such as sensation-seeking, urgency, premeditation, and perseverance, which are conceptually related phenotypes commonly used as measures of impulsivity [19]. Despite trait impulsivity being conceptualized as a rather static psychological construct, impulsive tendencies are not stable within individuals and may vary across different time points depending on the *state* of the subject, such as the current mood, level of anxiety or anger, and urges and desires as well as short-term situational

context [20, 21]. Unlike trait impulsivity, state impulsivity refers to impulsive behavior, which fluctuates based on the current state of the individual, measured with neurocognitive tasks and laboratory behavioral paradigms. The multidimensional nature of state impulsivity is evident from translational studies that find separate constructs of impulsive choice, impulsive action and self-reported impulsivity [5], revealing the non-unitary nature of impulsivity on a behavioral and pharmacological level [22]. Although state and trait impulsivity reflect several common and overlapping subconstructs that are often used interchangeably, they are theoretically dissociable and distinct in terms of etiology, conceptualization, and operationalization [23, 24]. In this respect, the use of both questionnaires and multiple behavioral measures – with behavioral measures having some advantages compared to self-report questionnaires, such as objectivity and less influence from past life events – may help researchers to discriminate between the stable personality characteristic 'trait impulsivity', and 'state impulsivity' which is activated by environmental factors and ever-changing contexts [18].

Numerous pieces of evidence show that diverse forms of impulsivity, measured by different tools and methods, are linked to drug-seeking and drug-taking behaviors. Multiple neurocognitive deficits – that may be either pre-existing and/or drug-induced – could differentially contribute to SUD-related processes [25]. For instance, impulsivity is thought to act as both a risk factor and a consequence of substance use, showing a bi-directional relationship with addictive behaviors [26, 28]. More specifically, according to several longitudinal and cross-sectional studies, more impulsive youth (as measured by self-report measures of trait impulsivity) are more likely to experiment with drugs, use drugs regularly and develop SUD later in life [27, 28]. Conversely, impulsive tendencies may also be a consequence of acute or chronic substance use. Consumption of substances with abuse potential has been shown to be capable of modulating state impulsivity (i.e., impulsive choice and action), albeit the magnitude and direction of effect appear contingent on baseline functioning of impulsivity [29].

Overall, higher levels of impulsivity are associated with SUD severity, higher rates of relapse vulnerability [30], and shorter duration of abstinence [31] across various substances of abuse. Impulsivity also appears to be a key predictor of substance use treatment outcomes which warrants more attention in the improvement of SUD treatment strategies to gain optimal outcomes [32]. Although the probability of relapse is increased by both trait and state impulsivity [33], state impulsivity is changeable and may provide some degrees of freedom for controlling impulsive choices. Empirical findings suggest that behavioral measures of impulsive choice and impulsive action that measure state impulsivity have a predictive value for treatment success in SUD patients, which can inform neurocognitive

and pharmacological enhancement and enable treatments tailored to individual patients in different states and contexts [34, 35]. In recent years, there has been a growing interest in exploring the state component of impulsivity, its neurocognitive mechanisms, and the relevant interventions that could benefit addiction clinical practice. The aim of this review is to provide an update on recent findings that identify neurocognitive mechanisms related to impulsivity in the context of substance use. We will further discuss state impulsivity as one of the objective indices of the intraindividual variability in impulsivity that varies across situations [36]. We also present a conceptual framework depicting how several neurocognitive mechanisms may shape impulsive behaviors – a proxy for SUD-related maladaptive behaviors – harnessing the cue-reactivity paradigm. In the last part of this paper, we will briefly review the interventions that could potentially reduce state impulsivity in people with SUD.

Method

We conducted a narrative overview of the existing literature on impulsivity, focusing on the state aspect that may result in momentary increases in impulsive behaviors, specifically related to choosing and taking drugs. We searched PubMed to provide a recent update on this topic, particularly articles published within the last five years, highlighting the neurocognitive processes involved in state impulsivity. We also included seminal studies to provide foundational knowledge and findings on the neural networks and cognitive processes shaping Context-dependent impulsivity (CDI). The search keywords included impulsivity, trait impulsivity, state impulsivity, substance use disorder, drug cue-reactivity, and neurocognitive mechanisms.

State Impulsivity as a Cognitive Marker of Cue-Reactivity

As mentioned earlier, state impulsivity is transient and varies over time within individuals [37]. Typically, state impulsivity emerges with little conscious awareness in response to a specific cue in the heat of the moment [38]. Such cues have specific salient properties (rewarding/reinforcing) that could induce high-arousal emotional states in terms of both negative (e.g., anxious) and positive (e.g., hedonic) valence or activate automatic approach tendencies [39]. Therefore, in addition to physiological changes such as heart rate, sweat gland activity, and skin temperature in a person with SUD, cue exposure could evoke internal states that consequently stimulate specific behavioral schemas, herein impulsive drug-seeking or drug-taking actions. It should be noted that cues could originate from both the external and internal environments [40]. Externally driven cues are perceived

through perceptual systems (e.g., visual, auditory, tactile, olfactory), whereas internal cues are generated from stored memories, interoception or mental imagery representations. Although drug-associated memories and imagery could be involuntarily evoked, they typically require a contextual cue. Thus, regardless of whether they are internal or external, contextual cues elicit a sudden motivational desire to pursue rewards by increasing pleasure, relieving stress/negative affects, as well as meeting homeostatic hedonic or allostatic needs. The latter occurs when individuals are in poor condition and are unable to meet their biopsychosocial needs if they only take low-risk options. Consequently, they are required to make risky decisions in order to obtain desirable outcomes [41]. For example, individuals who feel desperate and perceive that nothing can help them deal with stressful conditions may view drug use as the risky option that offers the only possibility of meeting their need to cope with stress and reduce tension. A temporary disadvantage (e.g., feeling desperate) favors temporarily escalated risk-taking (e.g., using drugs). The state impulsivity which is evoked by contextual cues, can be termed as CDI, which not only varies from person to person but also within the same person over time. These changes could be monitored and assessed using cue-reactivity paradigms in combination with other measures of state impulsivity.

The cue-reactivity paradigm is a widely adopted means to study how people with SUD respond to contextual drug-related stimuli across various sensory modalities (e.g., vision, audition, touch, taste, and smell). The cue-reactivity paradigm applies previously learned stimuli that signal the availability of drugs, thus evoking an intense urge to seek and use drugs [42]. The cue-reactivity paradigm is a useful laboratory means to mimic how current or abstinent drug users will react toward drug-associated cues in their real lives [43]. Although cue-reactivity is a well-established laboratory method used to reflect physiological, expressive, behavioral, and neural responses associated with drug initiation, continuation, and relapse, there are some limitations, particularly regarding its clinical application within the addiction field. First, there is a lack of consistent and conclusive findings about the relationship between cue-reactivity and post-cue presentation clinical outcomes (e.g., relapse, craving). While some studies have found a link between clinical outcomes and cue reactivity, other studies have failed to report the same results [44]. Even in a few studies, participants did not respond to the presented cues or reported little craving [45, 46]. The inconsistent results in these studies might be due to the fluctuating nature of craving, which needs to be measured in real-time using continuous data collection methods such as EMA (Ecological Momentary Assessment). This ecologically validated approach repeatedly measures real-life behavior *in vivo*, representing more accurate fluctuations in craving status. Despite this

promising approach, only a minority of studies have used EMA as a craving indicator in response to cue exposure [47]. Second, the cue-reactivity paradigm may elicit brain activity across different brain regions involved in cognitive processes such as perception, attention, memory, reward, and emotion. In spite of this complex neural involvement, little evidence has been found to precisely decompose these cue-related activity into more elemental processes [42, 48]. In spite of all these controversies around cue-reactivity, this paradigm is still among the common methods within the addiction field, to assist clinicians and investigators in gaining better insight into recovery markers, including CDI, for drug users either before treatment onset or during the recovery process. Previous research has demonstrated that CDI is derived from the aberrant and exaggerated appetitive salience and failure in neurocognitive processes involved in self-control. Given the importance of identifying such processes, we aimed to propose a neuroscience-informed framework to elucidate the probable neurocognitive mechanisms that shape CDI and suggest potential interactions between them. This integrative approach could lead to theoretical and clinical advances in conceptualizing, measuring, and targeting CDI in individuals with SUD.

In the following sections, we discuss these probable neurocognitive mechanisms and how they might lead to relapse and poor treatment adherence, using the cue-reactivity paradigm as a challenge test to activate state impulsivity.

Context Dependent Impulsivity (CDI) and Relapse Vulnerability: Neuroscience Perspective

As argued in the last section, several studies have included impulsivity as an outcome or independent variable in addiction treatment, yet little attention has been devoted to the precise processes shaping this concept. Considering previous empirical and clinical studies, we can categorize the mechanisms related to CDI into four related but distinct mechanistic domains/processes: attention, reflection, decision-making, and motor skills. These mechanisms are known to indicate specific phenotypes of impulsivity, adopted from neuropsychological assessments and neurocognitive theoretical models. Within this field, impulsivity may particularly be fractionated into rapid response and choice impulsivity, and is also affected by other cognitive processes such as attentional control, reflection impulsivity, and decision-making [49, 50]. From the view of these theories, impulsive behaviors stem from the failure of top-down cortical systems to regulate prepotent motor or reward-driven responses [26].

Attentional Impulsivity In general, attentional impulsivity refers to the impaired ability to efficiently process incoming information, continuously focus on task-relevant information, and effectively filter out distracting information [14,

51]. As a personality trait, elevated attentional impulsivity measured by the Barratt Impulsiveness Scale (BIS) indicates poor ability to focus and cognitive instability [52]. In spite of the small effect size reported for the association between attention bias and impulsivity [39, 53], attentional impulsivity could be evoked through exposure to drug-related contextual cues. Such exposure could activate automatic cognitive processes (system 1) and shift attention towards drug cues while weakening cognitive control processes (system 2) [54]. This involuntary and swift allocation of attention results in preferential processing of drug cues, thereby debilitating individuals' ability to reorient their attention towards neutral stimuli. The impact of attentional bias has been well-studied in terms of increasing self-reported craving and contributing to maladaptive behaviors, mainly relapse [55–57]. Therefore, to measure context-dependent attentional impulsivity, tasks such as the drug Stroop task and visual probe task, which induce attentional bias through cue exposure, provide valuable data in laboratory settings [58]. Findings from neuroimaging studies identify the activation of the ventral attention network (VAN) with increasing attentional bias towards drug cues. This network involves brain regions, including the temporoparietal junction (TPJ), insula, and inferior frontal gyrus [59, 60] which are correlated with the involuntary orientation of attention toward salient stimuli [61].

Reflection Impulsivity Introduced by Kagan [62], impulsivity could be characterized by a lack of reflection to gather and analyze information and consider all the available alternatives before making a decision [63]. Reflective individuals tend to slow down decision-making processes to increase the accuracy of the ultimate decision, while those with inadequate reflection make intuitive and rash choices or are easily convinced at a point of relative uncertainty with minimal information [63, 64]. Reflective processes are not only limited to analyzing the incoming information from one's external world but are also applied to internally generated data from the visceral organs, such as heart and stomach. These internal signals convey important information about the individual's homeostasis and bodily arousal. The ability to perceive, interpret, and monitor this information depends on interoceptive ability, which is found to be disrupted in people with SUD [65]. Although impulsivity and interoception appear to be connected, the underlying mechanism remains unknown [65]. Implicit in this evidence is the assumption that reflection impulsivity affects the choice of action a person with SUD takes in response to inaccurate and undifferentiated bodily signals. This choice of action involves using drugs to immediately alleviate negative feelings [66]. For example, an increased heart rate may result from stressful emotional conditions, pain, or physical exercise, but for people with SUD, who tend to negatively interpret ambiguous

bodily sensations, this internal signal could be erroneously perceived as aversive stimuli (e.g., stress or anxiety), leading them to use drugs as a rash solution to resolve these feelings [65]. They may be unable to withhold their intuitive response for a more detailed interpretation of their interoceptive state. Using fMRI data, it is believed that the Default Mode Network (DMN), which includes regions such as the cingulate cortex, hippocampus, medial prefrontal cortex, and inferior parietal cortices [67], is a critical hub for various interoceptive functions and metacognition [68]. Tasks like the Matching Familiar Figures test (MFF), Cognitive Reflection Test (CRT) and the Information Sampling Task (IST) have been used to assess reflective aspects of impulsivity in substance users [8, 63, 69].

Choice Impulsivity Increased sensitivity to salient and immediate rewarding stimuli, as well as a lack of planning and forethought, are the core characteristics of impulsive decision-making [70], considered a prototypical example of choice impulsivity. Choice refers to the preference for smaller immediate rewards over larger delayed rewards (i.e., reward-seeking choices) and choices reflecting reduced concern about the long-term consequences of one's choices, along with reduced flexibility to change behavior based on past experiences and future goals (i.e., risky decisions)[11, 23, 71]. In the context of addiction treatment, context-dependent choice impulsivity hampers efforts to maintain abstinence, as it is strongly associated with delay discounting. This means that drug users are characterized by increased sensitivity to proximate rewards (immediate pleasure of drug taking) while having a blunted response to delayed or distal consequences (such as increasing the length of abstinence). Common tasks to measure choice impulsivity include the Pictorial Choice Task (PCT) [72, 73], gambling tasks such as the Iowa Gambling Task and the Game of Dice Task (choice under uncertainty) [74], risk-taking tasks such as the Balloon Analog Risk Task (BART) (risky choice), and delay discounting tasks (intertemporal choice) [71, 75]. Neuroimaging studies investigating neural correlates of impulsive decision-making have identified a reward-related network known as the salience network (SN), involved in identifying salient stimuli and incorporating this information to inform decision-making [76]. Altered activity in regions of the SN, including the anterior insula, dorsal anterior cingulate, and inferior parietal cortex, is associated with a strong reaction to salient drug-related stimuli in drug users, as well as increased craving and urge to use drugs [77].

Action Impulsivity (a.k.a. “rapid-response impulsivity”) Another type of impulsivity is commonly applied to actions, which is known as motor impulsivity. As a trait-like impulsivity, motor impulsivity is characterized

by acting without thinking and is measured with the BIS questionnaire. As a context-dependent state, motor impulsivity is linked to disinhibition, which refers to the inability to withhold a prepotent response (i.e., action inhibition) or interrupt an already initiated action (i.e., action cancellation) [14, 78, 79]. Similar to other forms of CDI, motor impulsivity could be a momentary response triggered by drug-related cues, and drug users often fail to override such an intense tendency toward acting upon a drug or drug cue following exposure. Motor impulsivity is reflected in terms of difficulty canceling or stopping a prepotent or already initiated action in experimental tasks such as go/no-go or stop-signal tasks [80, 81]. Integrating cue reactivity paradigms with these conventional tasks reveals altered activation in brain regions including the inferior frontal gyrus (IFG), anterior cingulate cortex (ACC), and dorsolateral prefrontal cortex (dLPFC), compromising the fronto-parietal network (FPN), also known as the executive control network (ECN). The FPN is correlated with goal-directed behaviors and response inhibition, and it is activated when individuals attempt to put a brake on their actions, similar to what happens when drug users try to abstain in response to cue exposure [61].

To summarize this section, four plausible neurocognitive mechanisms may be involved when individuals fail to control cognitive resources in favor of balancing between systems 1 and 2 thinking, particularly in response to contextual drug-related cues activating state impulsivity. The multidimensional nature of impulsivity is linked to the engagement of distinct brain networks and their interactions, trading off between controlled and impulsive responses. This interplay between the four neurocognitive mechanisms is demonstrated in a conceptual framework proposed to conceptually link neural, cognitive, and behavioral mechanisms underlying context-dependent state impulsivity (Fig. 1).

In the next section, we map this framework into the clinical context of addiction and discuss how neuroscience-informed interventions could target each mechanism to favor clinical outcomes of addiction treatment.

Targeting Context-Dependent Impulsivity Through Cue-Based Interventions

Due to the involvement of several cognitive mechanisms in shaping CDI, several neuroscience-informed interventions have been suggested to be translated into clinical practice for addiction treatment. In this section, we provide a brief overview of the interventions that could potentially help improve addiction treatment outcomes by influencing the underlying cognitive mechanisms.

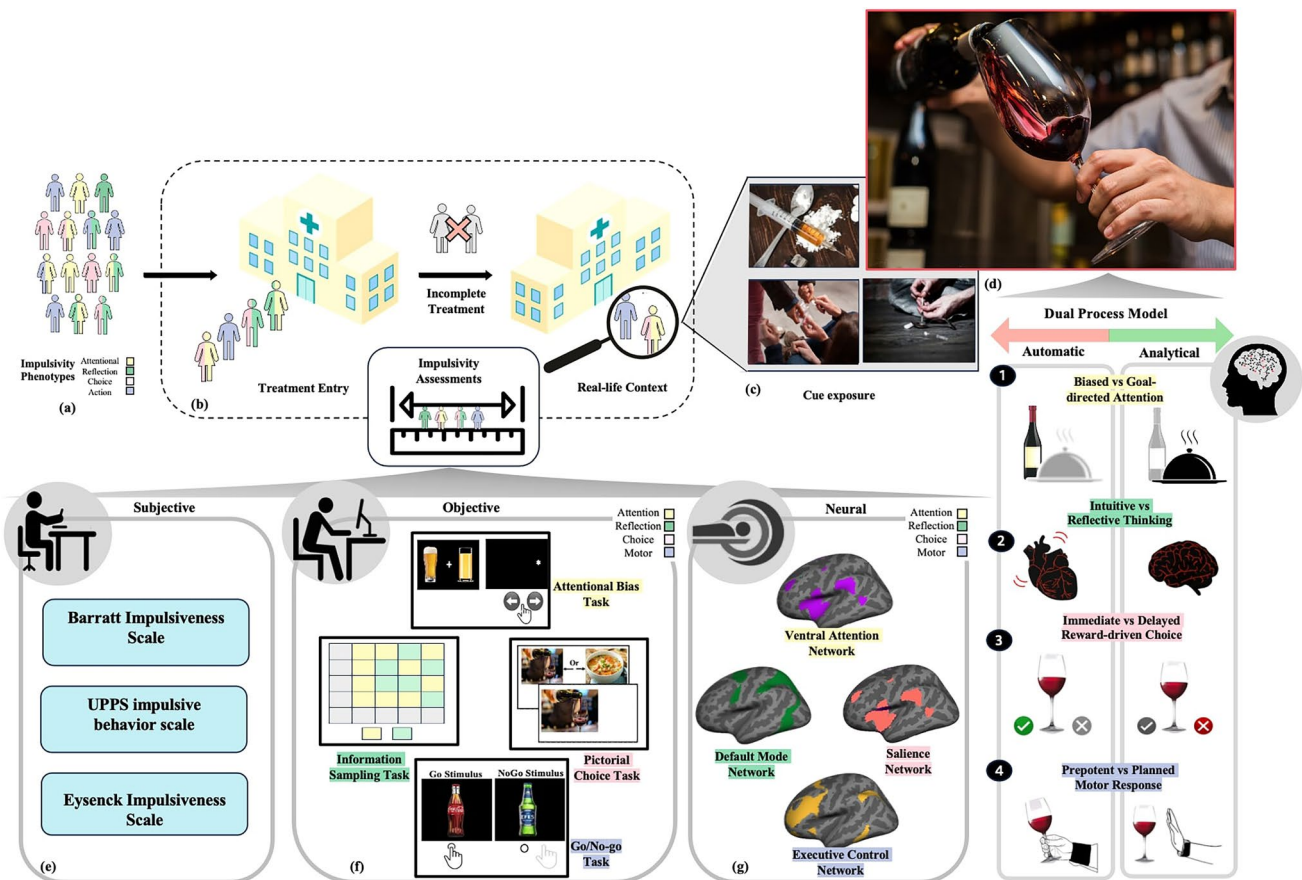


Fig. 1 Conceptual multilevel framework of context-dependent impulsivity. **a** different impulsivity transdiagnostic phenotypes of impulsivity (Attention, Reflection, Choice, and Action) and transdiagnostic phenotype; **b** the dimensional biobehavioral nature of impulsivity crosses diagnostic boundaries and makes treatment adherence and success rather challenging due to differences in symptom profiles of individuals. This challenge necessitates precise assessment tools for measuring impulsivity with high ecological validity, generalizable to real-life contexts. **c** In these real-life contexts, individuals may be exposed to drug-related cues, activating cognitive systems (1 & 2) that process stimuli differently. **d** These cognitive processes include attention, reflective processes, decision-making, and motor skills. Valid assessment tools and techniques for these processes comprise **(e)** subjective measures of impulsivity that include self-report scales and questionnaires (e.g., Barratt Impulsiveness Scale, UPPS Impulsive Behavior Scale), **(f)** objective measures of impulsivity that

include various cognitive tasks such as Attentional Bias Task (measures attention impulsivity by measuring attentional bias towards drug cues), Information Sampling Task (measures reflection impulsivity by requiring reflective rather than intuitive processes), Pictorial Choice Task (measures choice impulsivity by allowing individuals to freely choose to enlarge thumbnail pictures of either drug or non-drug by pressing a left or right arrow key), and Go/no-Go Task (measures motor impulsivity by requiring the withholding of a prepotent motor response). **g** Neural measures of impulsivity also map onto several networks in the brain, such as the Ventral Attention Network (related to attention impulsivity), Default Mode Network (related to reflection impulsivity), Salience Network (related to choice impulsivity), and Executive Control Network (related to motor impulsivity). UPPS: Urgency-Premeditation-Perseverance-Sensation Seeking-Positive Urgency. Figure created with Microsoft PowerPoint

Attentional Impulsivity To overcome and reduce attentional bias toward drug-related cues, Attentional Bias Modification (ABM) is used as a conventional intervention [82]. In this paradigm, individuals are simultaneously exposed to a drug-related cue and a neutral object. Throughout the training trials, individuals practice ignoring the stimulus containing drug-related information while reacting to the drug-irrelevant stimulus immediately [58, 83]. Various forms of ABM, such as visual probe and Drug Attention Control Training Program (ACTP), have been used for different groups of

substance users with mixed outcomes in terms of altered attentional bias and addiction-related symptoms (e.g., abstinence, craving, number of lapses) [58, 83]. Another promising intervention to control attention bias is cognitive reappraisal, which operates through regulating emotion [84, 85]. By using cognitive reappraisal in response to drug-related cues, individuals practice reinterpretation of the meaning of the stimuli in order to reduce their craving. For example, when seeing a picture of cocaine, they change their

perspective by thinking that the cocaine is not real or that it is just flour [55].

Reflection Impulsivity To improve interoception, individuals are required to enhance their awareness of internal bodily sensations and related process of metacognitive awareness [86]. Thus, interventions aimed at facilitating this awareness could be beneficial for the treatment of addiction. Through practicing mindfulness-based trainings, individuals gradually learn how to smoothly switch their attention from the external world to internal sensations and improve their sustained present-moment internal awareness non-judgmentally. In the training sessions, individuals learn to become observers of their unpleasant thoughts, feelings, and physical sensations without trying to change, escape, or avoid them. Thus, for people with SUD, specific mindfulness-based interventions (e.g., Mindfulness-Based Relapse Prevention, Mindfulness-Based Addiction Treatment) have been developed in combination with cue exposure. During the training sessions, individuals visualize triggers or high-risk situations to generate craving and then attempt to deal with this negative experience through observation, experiencing, and accepting without reacting [87]. Previous studies have found that through consistent mindfulness practice, drug users become more able to regulate their emotions and prevent relapse [86, 88].

Choice Impulsivity Impaired decision-making is characterized by a tendency toward immediate rewards and risk-taking in drug users. One of the key mechanisms underlying this phenotype is the narrowed temporal window and lack of consideration of long-term consequences among users [89]. Assigning incentive value to a drug cue makes users less sensitive to the probable outcomes of using drugs and thus devalues the delayed rewards of staying abstinent in the long term. Interventions that expand temporal horizons and promote future thinking have been found to be useful in reducing delay discounting [90–92]. Episodic future thinking has recently been informed by cue exposure paradigms as well as memory reconsolidation [93]. Through this process, individuals are presented with a personalized drug cue or invited to imagine themselves in a hypothetical drug-related situation based on their personal experiences (e.g., being offered drugs by peers). After this exposure, their maladaptive drug-related memories become activated, and they are then guided to imagine a similar scenario happening in the future. Once the image of this high-risk situation is vividly generated, individuals are asked to elaborate on its episodic details, mentally predict the alternatives to choose, and anticipate their consequent outcomes. The last two stages involve intention and planning their desirable future action. The potential outcome of this process would result in an updated script of past drug-related memories re-encoded

with abstinence-oriented values. This is a novel approach that requires more evidence-based investigation to be validated and translated into clinical practice.

Action Impulsivity A well-studied set of interventions to reduce motor impulsivity among drug users is inhibitory control training (ICT) programs, in which individuals are instructed to repeatedly withhold or cancel their motor response to drug-related stimuli while continuing to press a key upon presentation of drug-irrelevant stimuli [94]. This paradigm could gradually improve top-down inhibitory control over bottom-up impulsive actions. Similar to assessments, the Go/NoGo (GNG) and the Stop-Signal paradigms are the two most commonly used interventions to train individuals to gain control over their motor impulsivity, delivered via single or multiple training sessions [94–96].

Together, using our conceptual framework could illuminate different aspects of context-dependent impulsivity, which are considered as a marker of cue reactivity. Findings from previous studies show promising futures for interventions that embed cue exposure into conventional behavioral interventions, particularly cognitive training. This new generation of interventions could improve the generalizability of in vitro studies to real-world contexts where substance users may repeatedly encounter high-risk situations.

Discussion

The purpose of this paper is to provide insight into the neurocognitive mechanisms that likely shape different phenotypes of context-dependent impulsivity, specifically in terms of attention, reflection, choice, and action.

Our main assumption in this paper is based on the dual process model, which posits that self-control failures stem from an imbalance between the two thinking systems (1 & 2) in people with substance use disorder. Broadly speaking, a marked shift of power from system 2 (characterized by controlled, reflective, slower deliberative processes) to system 1 (characterized by impulsive, reflexive, automatic, spontaneous processes) defines substance users as impulsive individuals [9]. These individuals act quickly, with a diminished ability or willingness to consider the consequences of their actions. From a neuroscience perspective, impulsive responses may be elicited by salient, personally relevant incentive stimuli (such as drug-related cues or memories) [97], suggesting the concept of context-dependent impulsivity. Consistent with previous theoretical and empirical studies, we adopted four key cognitive mechanisms likely influenced and heightened by the activation of System 1 and the impulsive state. These four distinct but interrelated

mechanisms include attentional control, reflective thinking, decision-making, and motor control. Each of these mechanisms may be disrupted in association with high impulsivity, failing to maintain their goal-oriented and controlled functions. As a result, attentional bias, intuitive choices, risky decision-making, and rash, appetitive prepotent motor actions tend to emerge in substance users in response to contextual drug-related cues. Focusing on evidence from addiction research, we propose a multi-level conceptual framework that integrates findings from classic neurocognitive measures and the cue-reactivity paradigm to explain these mechanisms.

According to our framework, state impulsivity can be considered in terms of four phenotypes, each of which is investigated at multiple levels of analysis, including neural, cognitive, and behavioral. This new insight could provide a means to increase the precision of targeting these mechanisms by performing comprehensive assessments and then personalizing therapeutic interventions in clinical practice, potentially leading to improved treatment outcomes. In the following sections, we delve into this idea in more detail.

Cognitive Markers of Impulsivity Before initiating treatment, therapists could benefit from using a multi-dimensional assessment that measures various types of impulsivities (attentional, reflection, action, and choice), ideally administered longitudinally (e.g., via ecological momentary assessment), to gain better insight into the most disrupted neurocognitive mechanism(s) and determine if these mechanisms are context-dependent. With this personalized profile, patients are identified and monitored during recovery based on their specific biobehavioral markers of state impulsivity. As discussed earlier, cues and craving are critical ingredients in the cycle of drug seeking and relapse. Therefore, embedding cue-reactivity paradigms within standard neuropsychological assessments could provide more real-world data regarding the specific needs of patients. Moreover, the cautious application of personalized cues (drug-related, traumatic, emotional cues) needs to be taken into careful consideration to evoke state impulsivity that resembles real-life situations.

Tailoring Treatment Based on Patient's Profile Based on the individual patient's profile on different dimensions of context-dependent impulsivity, therapeutic interventions could be planned and continuously monitored for possible changes in the target mechanism(s). For example, for a patient with heightened attention to drug-related cues, neurocognitive interventions could be more beneficial if they include attentional bias modification. For individuals exhibiting impairments in decision-making, episodic future thinking training

could expand the temporal window, gradually decreasing delay discounting.

Overall, we suggest that considering state impulsivity as a multi-dimensional construct, with various underlying mechanisms measured through baseline neurocognitive assessments prior to initiating any treatment for people with SUD, may yield substantial benefits. Tailoring interventions to meet the specific needs of patients increases the likelihood of treatment retention and greater engagement with treatment materials, facilitating their practical application in real-life [98].

Conclusion

In conclusion, context-dependent state impulsivity can be considered a recovery marker in people with SUD, using the cue-reactivity approach alongside other measurements. Variation in this transdiagnostic phenomenon may be linked to four cognitive processes: attention, reflection, action, and choice. Although these mechanisms are related to the general concept of impulsivity, they could be specifically influenced in the context of drug cue exposure. Each of these mechanisms is associated with specific neural and behavioral measures and can be assessed and modulated using various neurobehavioral tasks and interventions.

Our conceptual framework could be validated in future studies through the implementation of a comprehensive and ecologically validated assessment at various levels of analysis. This multilevel analysis could then offer a phenotypic framework for clinicians and researchers to classify and diagnose individuals with SUD, providing personalized interventions [98].

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Through coordinate-based meta-analyses, authors examined the common and distinct brain activity convergence in cue-reactivity studies involving drug and natural visual stimuli.

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The current study investigated the effect of receiving adjunct cognitive rehabilitation and baseline executive

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This study introduces the Cognitive Impulsivity Suite, comprising three computerized/online tasks using a gamified interface.

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This systematic review sought to evaluate the efficacy of mindfulness-based relapse prevention (MBRP) for individuals suffering from substance use disorders (SUD), with a focus on relevant clinical variables.

- Rafei P, Rezapour T, Batouli SAH, Verdejo-García A, Lorenzetti V, Hatami J. How do cannabis users mentally travel in time? Evidence from an fMRI study of episodic future thinking. *Psychopharmacology.* 2022;239(5):1441–57. <https://doi.org/10.1007/s00213-021-06002-7>.

Authors in this study aimed to compare brain activations of regular cannabis users and non-using controls during an EFT fMRI task and investigate the association between EFT and cannabis use variables.

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This study assesses the feasibility of utilizing EFT methods in individuals with cocaine use disorder and conducts a preliminary measurement of the EFT effect on discounting behavior in this population.

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This study proposes a new framework as a cognitive intervention termed Cue-induced Retrieval and Reconsolidation with Episodic Foresight (CIREF) for combining episodic future thinking with cue-induced memory reconsolidation.

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In this study, authors aimed to identify whether individuals with tendencies towards behavioral addictions show stimulus-specific reductions in inhibitory control.

- Vassileva J, Lee J-H, Psederska E, Ahn W-Y. Utility of Computational Approaches for Precision Psychiatry: Applications to Substance Use Disorders. In: Stoyanov D, Draganski B, Brambilla P, Lamm C, editors. *Computational Neuroscience*. New York, NY: Springer US; 2023. p. 211–31.

In this chapter, the authors advocate for the potential of theory-driven and data-driven computational approaches to significantly enhance the precision and reliability of measurement and the accuracy of diagnosis and prognosis in psychiatry.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

Competing Interest The authors declare no competing interests.

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